

**HOW TO USE THIS SET**

20 questions, mixed difficulty, 5 case-based. Attempt every question before checking the answer key on the last page.

● 4 Easy

● 10 Medium

● 6 Hard

1.**EASY**

Which hepatitis virus is transmitted by the fecal-oral route and does not cause chronic infection?

- A. All of the above
- B. Hepatitis A virus
- C. Hepatitis D virus
- D. Hepatitis C virus
- E. Hepatitis B virus

2.**EASY**

Chronic hepatitis is defined as evidence of continuing or relapsing hepatic disease for more than:

- A. 3 months
- B. 1 year
- C. 6 months
- D. 1 month
- E. 2 weeks

3.**EASY**

Hemochromatosis is how many times more common in males than females?

- A. 2 times
- B. 10 times
- C. 20 times
- D. 5 times
- E. Equal frequency

4.**EASY**

Which gene is mutated in Wilson disease?

- A. ATP7B
- B. HFE
- C. BRCA1
- D. SERPINA1
- E. APC

5.**MEDIUM**

Which hepatitis virus requires co-infection or superinfection with HBV to replicate?

- A. Hepatitis C virus
- B. Hepatitis G virus
- C. Hepatitis D virus
- D. Hepatitis A virus
- E. Hepatitis E virus

6.**MEDIUM**

On hepatitis B serology, which marker's continued presence indicates a carrier state?

- A. HBeAb
- B. HBsAg
- C. Anti-HBs
- D. Anti-HCV
- E. Anti-HAV IgM

7.

MEDIUM

Which histologic feature distinguishes chronic active hepatitis from chronic persistent hepatitis?

- A. Presence of Kupffer cells
- B. Inflammation spilling into the parenchyma (interface hepatitis)
- C. Presence of any portal inflammation
- D. Absence of fibrosis
- E. Presence of granulomas

8.

MEDIUM

CASE

A patient who takes acetaminophen in excess develops predictable, dose-dependent liver necrosis.

This mechanism of drug-induced liver injury is best classified as:

- A. A carrier state
- B. An autoimmune reaction
- C. A predictable reaction
- D. A vascular seeding infection
- E. An idiosyncratic reaction

9.

MEDIUM

Liver flukes are a recognized predisposing factor for which tumor, particularly in Southeast Asia?

- A. Hepatocellular adenoma
- B. Hepatocellular carcinoma
- C. Angiosarcoma
- D. Neuroendocrine tumor
- E. Cholangiocarcinoma

10.

MEDIUM

Hepatic steatosis occurs in what proportion of chronic excessive alcohol consumers?

- A. 90-100%
- B. 50-60%
- C. 10-35%
- D. 8-20%
- E. 1-5%

11.

MEDIUM

NAFLD is associated with which cluster of findings?

- A. Type 2 diabetes, central obesity, dyslipidemia, hypertension
- B. Type 1 diabetes and low BMI
- C. Low ceruloplasmin and Kayser-Fleischer rings
- D. Chronic alcohol use
- E. Iron overload and skin bronzing

12.

MEDIUM

Which mechanism explains liver fibrosis in hemochromatosis?

- A. Copper deposition in Kupffer cells
- B. Alcohol-induced neutrophilic inflammation
- C. Viral cytopathic injury
- D. Iron-induced activation of hepatic stellate cells stimulating collagen formation
- E. Autoimmune destruction of hepatocytes

13.

MEDIUM

Steatohepatitis in chronic alcohol use is characterized histologically by which type of inflammatory infiltrate?

- A. Lymphocytic infiltration only
- B. Neutrophilic infiltration
- C. Granulomatous inflammation
- D. Plasma cell-rich infiltration
- E. Eosinophilic infiltration

14.

MEDIUM

Which extrahepatic organ is affected in hemochromatosis leading to hypogonadism?

- A. Parathyroid glands
- B. Thyroid
- C. Adrenal glands
- D. Pituitary only
- E. Gonads

15.

HARD CASE

A 50-year-old with chronic hepatitis B develops progressive jaundice, ascites, and an elevated AFP. Imaging reveals a liver mass.

Which sequence best reflects the natural history that led to this presentation?

- A. Chronic hepatitis leading to cirrhosis and/or hepatocellular carcinoma
- B. Subclinical disease leading to cirrhosis without an intervening hepatitis phase
- C. Autoimmune hepatitis superimposed on hepatitis A
- D. Acute hepatitis leading directly to fulminant hepatitis
- E. Carrier state leading directly to hepatocellular carcinoma without hepatitis

16.

HARD CASE

A liver biopsy shows dense portal mononuclear infiltrate, interface hepatitis, bridging fibrosis, and ground-glass hepatocytes.

Which diagnosis best fits this biopsy?

- A. Wilson disease
- B. Acute hepatitis A
- C. Chronic hepatitis B
- D. Alpha-1-antitrypsin deficiency
- E. Alcoholic hepatitis

17.

HARD CASE

A patient with cirrhosis and a history of decades of heavy alcohol use achieves complete abstinence.

Which outcome is most accurate regarding potential regression of their liver disease?

- A. Cirrhosis always fully reverses with abstinence
- B. Abstinence has no effect once cirrhosis is established
- C. Partial regression of scarring and a shift from micronodular to macronodular cirrhosis may occur, but complete regression is rare
- D. Abstinence only prevents further alcohol-related organ damage outside the liver
- E. Cirrhosis converts to hepatocellular adenoma with abstinence

18.

HARD

Which combination of laboratory findings best supports a diagnosis of Wilson disease over hemochromatosis?

- A. Elevated serum iron and ferritin with normal ceruloplasmin
- B. Normal ceruloplasmin with elevated hepatic iron
- C. Elevated serum copper with normal ceruloplasmin
- D. Low serum ceruloplasmin with elevated hepatic copper
- E. Elevated transferrin saturation with low hepatic iron

19.

HARD CASE

A young adult with cirrhosis and a family history of early-onset emphysema undergoes liver biopsy showing PAS-positive, diastase-resistant globules in hepatocytes.

The mechanism of hepatocyte injury in this condition is best explained by:

- A. Iron-catalyzed free radical lipid peroxidation
- B. Accumulation of misfolded protein causing endoplasmic reticulum stress and apoptosis
- C. Unopposed neutrophil elastase activity destroying hepatocytes
- D. Copper-induced free radical injury
- E. Autoantibody-mediated hepatocyte destruction

20.

HARD

A patient with chronic HCV infection is counseled about long-term risk. Which statement best reflects the natural history described for HCV?

- A.** Fewer than 10% develop chronic infection
- B.** HCV never causes hepatocellular carcinoma
- C.** Cirrhosis develops in the majority within 1 year
- D.** Acute HCV is typically more severe than acute HBV
- E.** 80-90% develop persistent infection/chronic hepatitis, and about 20% of those progress to cirrhosis

ANSWER KEY & RATIONALE

Don't peek until you've committed to an answer on every question.

Q	Ans	Why
1	B	HAV (like HEV) is transmitted fecal-orally and does not cause chronic infection or a carrier state.
2	C	Chronic hepatitis is defined as symptomatic, biochemical, or serologic evidence of continuing/relapsing disease for more than 6 months.
3	D	Hemochromatosis affects about 5 times as many males as females.
4	A	Wilson disease is caused by loss-of-function mutation of ATP7B on chromosome 13.
5	C	Hepatitis D virus is a defective virus that requires HBsAg and replicates only in HBV-infected cells.
6	B	HBsAg is positive during acute disease, and its continued presence indicates the carrier state.
7	B	In chronic active hepatitis, inflammation spills into the parenchyma (interface hepatitis), whereas in chronic persistent hepatitis it remains confined to the portal tracts.
8	C	Acetaminophen toxicity is the classic example of a predictable, dose-dependent drug reaction.
9	E	Liver flukes are a well-recognized predisposing factor for cholangiocarcinoma, particularly in Southeast Asia.
10	A	Hepatic steatosis occurs in 90-100% of chronic excessive alcohol consumers, the most common early stage of alcohol-related liver disease.
11	A	NAFLD develops in association with the metabolic syndrome: type 2 DM, central obesity, dyslipidemia, and hypertension.
12	D	Excess iron activates hepatic stellate cells, stimulating collagen/fibrosis formation, in addition to causing lipid peroxidation and DNA damage.
13	B	Alcohol-related steatohepatitis is characterized by neutrophilic infiltration, driven by further fat accumulation and the toxic effect of ethanol.
14	E	Iron deposition in the gonads causes hypogonadism in hemochromatosis, alongside skin bronzing, secondary diabetes, and cardiac disease.
15	A	Chronic HBV infection can progress to cirrhosis and/or hepatocellular carcinoma, which interact bidirectionally (cirrhosis increases HCC risk and vice versa can coexist).
16	C	Ground-glass hepatocytes together with chronic hepatitis features (interface hepatitis, bridging fibrosis) are classic for chronic hepatitis B.
17	C	Even with established cirrhosis, complete abstinence may lead to partial regression of scarring and a shift from micronodular to macronodular cirrhosis; complete regression is rare.
18	D	Wilson disease is diagnosed by low serum ceruloplasmin and elevated hepatic copper content (with increased urinary copper); hemochromatosis instead shows elevated serum iron and ferritin.
19	B	In alpha-1-antitrypsin deficiency, liver damage results from accumulation of misfolded a1AT protein within hepatocytes, causing ER stress and apoptosis; the lung is instead damaged by unopposed elastase activity.
20	E	HCV causes persistent infection/chronic hepatitis in 80-90% of patients, with cirrhosis developing in about 20%; acute HCV illness is usually less severe than HBV.

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Recalled and source questions are treated as evidence of what gets asked, not as a source of correct answers — verify against your lecture notes.